

SurRF: Unsupervised Multi-View Stereopsis by Learning Surface Radiance Field

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Abstract—The recent success in supervised multi-view stereopsis (MVS) relies on the onerously collected real-world 3D data. While the latest differentiable rendering techniques enable unsupervised MVS, they are restricted to discretized (e.g., point cloud) or implicit geometric representation, suffering from either low integrity for a textureless region or less geometric details for complex scenes. In this paper, we propose SurRF, an unsupervised MVS pipeline by learning Surface Radiance Field, i.e., a radiance field defined on a continuous and explicit 2D surface. Our key insight is that, in a local region, the explicit surface can be gradually deformed from a continuous initialization along view-dependent camera rays by differentiable rendering. That enables us to define the radiance field only on a 2D deformable surface rather than in a dense volume of 3D space, leading to compact representation while maintaining complete shape and realistic texture for large-scale complex scenes. We experimentally demonstrate that the proposed SurRF produces competitive results over the state-of-the-art on various real-world challenging scenes, without any 3D supervision. Moreover, SurRF shows great potential in owning the joint advantages of mesh (scene manipulation), continuous surface (high geometric resolution), and radiance field (realistic rendering).

Index Terms—Multi-view stereopsis, unsupervised learning, neural rendering

1 INTRODUCTION

MULTI-VIEW stereopsis (MVS) aims to recover dense surface in 3D space from a set of 2D images with known camera parameters. Compared with the traditional methods [1], [2], recent learning-based works [3], [4], [5], [6], [7] leverage learned geometry and illumination priors to obtain more robust and complete reconstruction. However, all of them require onerous laser-scanned 3D data for training, which is not always available in outdoor scenes.

To overcome this barrier, some unsupervised MVS methods [8], [9] propose reprojection losses to encourage cross-view

photo-consistency. However, they are susceptible to texture patterns and environmental illuminations, lead to incomplete 3D reconstructions around textureless regions. Besides, they are restricted to specific 3D representations (e.g., voxel and point cloud) that tend to lose smooth and continuous manifold structure for high fidelity surfaces [10], which suffer from discretization artifacts. While the continuous mapping defined on the mesh solves the problem of discretization, it requires a fixed geometric position, which cannot be used for geometry reconstruction. The other line of the solution [11], [12], [13] propose differentiable renderer to progressively update the geometry from the rendered images. Moreover, implicit representations for both shape and texture can be learned from multiple images with continuous water-tight mesh as output [11], [13], [14]. However, the radiance field, implicitly defined using a neural network, is redundantly distributed in a dense volume of 3D space. That severely limits the representation ability of the implicit function for detailed geometric content.

In this paper, we propose a novel unsupervised MVS pipeline by learning Surface Radiance Field (SurRF), a radiance field explicitly defined on continuous 2D surface, being optimized without 3D supervision. Our key insight is that, in a local region, the continuous surface can be explicitly defined as the shift along view-dependent camera rays without any 3D supervision. That enables us to learn the radiance field on a piece of a 2D deformable surface rather than in a dense volume in 3D space and makes the neural function more effective to represent a complex 3D scene.

Precisely, surface radiance field (SurRF) is a continuous representation of both explicit surface deformation w.r.t. an initialized coarse-level mesh and implicit radiance field, encoding geometry, texture, and lighting conditions. Specifically, for each point (u, v) along each direction (θ, ϕ) on a local triangle of the primary mesh, SurRF is a continuous

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function whose input is a 4D vector field (u, v, θ, ϕ) , and the output is the coordinate and color of the intersection of each ray through the target surface. This continuous function, constructed by the multi-layer perceptron (MLP), encodes the 2D surface deformation and the radiance field on the deformed 2D surface. To combine multiple local SurRFs into a complex global model, we use a differentiable rendering framework to average the nearest pieces of surface hit by each pixel ray. Since this process is fully differentiable, SurRF can be optimized through back-propagation without any 3D supervision by comparing the rendered images with the multi-view target images. Minimizing this error encourages multi-view photo-consistency because only when the point is on the actual surface will the MLP predict accurate color with few multi-view variations. To enable highly accurate and globally continuous surfaces in complex scenes, we first separate the color prediction with different appearances such as surface reflectance and shading. This implicit modeling of illumination enforces accurate geometry learned from the rendering equation. Furthermore, we propose a global shape continuity regularization to punish the misaligned geometry and texture predictions between nearby surfaces.

Our proposed SurRF significantly outperforms the state-of-the-art method of unsupervised MVS methods, including DTU[19] and Tanks and Temples[18], and produces competitive results over the recent supervised ones with high fidelity, resolution, and detail. Since SurRF is enabled to map continuously from a coarser triangle mesh to the finer geometry, the continuous mapping in the local region does not laboriously need to fit the geometry and the texture information in the dense volume space like Nerf[20] and IDR[13], leading to compact representation and globally maintaining complete shape for large-scale complex scenes. Besides, unlike the surface parameterization methods [10], [21] that optimize the geometry merely via a collection of parametric surface elements, our learned view-dependent function provides a neat solution from multi-view images to both the texture and the geometry processing, which is more generalized for outdoor scenes with complex illuminations. Overall, SurRF maintains the advantages of mesh (scene manipulation), continuous surface (high geometric resolution), and radiance field (realistic rendering).

2 RELATED WORK

2.1 Multi-View Stereopsis Reconstruction

Works in the multi-view stereopsis (MVS) field can be roughly categorized into 1) direct point cloud reconstructions 2) depth maps fusion algorithms, and 3) volumetric methods. Point-cloud-based methods operate directly on 3D points, which progressively densify the reconstruction in a propagation manner[22], [23]. As the propagation of point clouds proceeds sequentially, these methods are difficult to be fully parallelized and take a long time to process. Depth map fusion algorithms [1], [2], [23], [24], [25] decouple the complex MVS problem into view-wise depth map estimation and then fuse together with the point cloud[26], [27], [28], [29]. [24] deals with ultra-high-resolution image sets and uses a robust descriptor for efficient matching purposes. [23] describes a patch model that consists of a quasi-dense set of rectangular patches covering the surface. To aggregate image similarity across multiple views, [1]

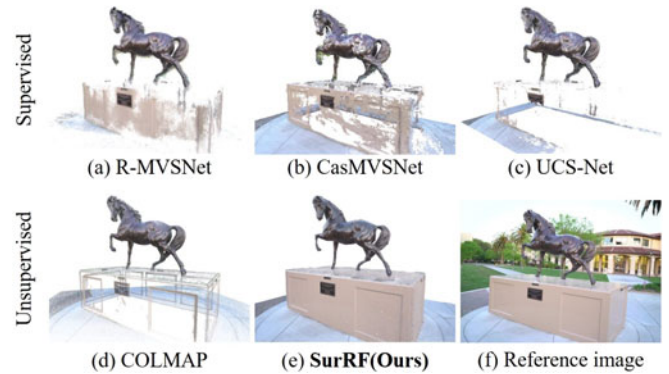


Fig. 1. Point cloud comparison with the state-of-the-art multi-view stereo approaches (a) R-MVSNet [4], (b) CasMVSNet [16], (c) UCS-Net [17], (d) COLMAP [15], (e) SurRF(Ours), (f) Reference image in Tanks and Temples dataset [18]. Our proposed SurRF generates much more complete, delicate, and realistic surface predictions.

obtains more accurate depth maps. After getting multiple depth maps, depth map fusion algorithms integrate them into a unified and augmented scene representation while mitigating any inconsistencies among individual estimates. To improve the fusion accuracy, [25] learns several sources of depth map outliers. [27] fuses multiple depth estimations into a surface by evaluating visibility in 3D space and attempts to reconstruct the region that is not directly supported by depth measurements. [29] proposes to target the reconstruction from crowd-sourced images explicitly. [28] proposes a variational depth map formulation that enables parallelized computation on the GPU. COLMAP[2] directly maximizes the estimated surface support in depth maps and allows dataset-wide, pixel-wise sampling for view selection. However, these algorithms suffer from incomplete fusion model with large baseline angle or occluded views since skewed patches and uneven samples on the 3D surface in these cases leads to poor quality photo consistency agreements.

Volumetric-based methods divide the 3D space into regular grids. They use either implicit representation[30], [31], [32] or explicit surface properties[5], [33], [34], [35] to represent and optimize in a global framework. To improve the reconstruction scalability, some methods either apply an octree representation[32], [36], or allow a hierarchical multi-scale structure[6], [37], [38]. However, the voxel point in these representations does not directly represent neighborhood information with shapes and colors, making it hard to generate high fidelity surfaces with realistic texture and smooth surface via unsupervised learning.

2.2 Learning-Based MVS

Many learning-based MVS methods demonstrate promising performance in recent years. 2D-convolutional neural networks (2D-CNNs)[39], [40], [41], [42] are applied to various stages in MVS. In [42], a CNN is trained to predict the normals of a given depth map based on image appearance. The estimated normals are then used to improve the depth map fusion. Compared to its previous work [1], the approach increases the completeness of the reconstruction at the cost of a slight decrease in accuracy. Deep learning has also been successfully applied to other 3D applications like volumetric shape retrieval or object classification [43], [44], [45], [46]. In order to simplify the retrieval and representation of 3D

shapes by CNNs, [47] introduces a representation, termed geometry image, which is a 2D representation that approximates a 3D shape. While the 3D reconstruction of an object from a single or multiple views using convolutional neural networks has been studied in [48], [49], the approaches focus on a very coarse reconstruction without geometric details when only one or very few images are available.

However, the performances of the previous methods are limited due to the lack of contextual geometry knowledge, so currently, learning methods based on 3D cost volume regularization have been proposed. By projecting 2D images or features into 3D volume, SurfaceNet[5], [6] optimizes the 3D geometry by 3D CNN, and the similar 3D cost volume regularization methods are proposed in [3], [4], [5], [6], [34], [50], [51]. However, the cost volume with uniformly sampled depth hypotheses often require large GPU memory and is hard to reconstruct large scale or high resolution scenes. To overcome such limitations, some methods[16], [17], [52] further improve the cost volume to a deformable one or point cloud structure for memory and efficiency consideration. Point-MVSNet[52] applies a small cost volume to generate the coarse depth and uses a point-based iterative refinement network to output the full resolution depth. CasMVSNet[16] and UCS-Net[17] both construct coarse-to-fine cost volume using adaptively varying depth hypothesis. PVA-MVSNet[51] proposes to generate pyramid depth maps and progressively refine the finer depth map with reliable depths from a coarser level. However, these supervised methods crucially rely on ground-truth 3D data. To realize unsupervised MVS, [8], [9], [53] replace the supervision signal with the unsupervised reprojection or cross-view consistency loss. However, reprojection under multiple views is highly sensitive to environmental illumination and self-occlusion, which is hard to generalize in complex scenes.

2.3 Neural Surface Representations

Some recent work also investigate the reconstruction of surface meshes[54], [55], [56], [57], [58], [59], [60], deformable shapes[55], [61] and implicit representations like learned distance functions[62], [63], [64] or occupancy fields[65], [66]. [11] represents surfaces as 3D occupancy fields and finds the surface intersection for each ray using a numerical method, then calculates an exact derivative using implicit differentiation. Each ray intersection location is provided as the input to a neural 3D texture field that predicts a diffuse color for that point. [12] uses a neural 3D representation that outputs a feature vector and RGB color at each continuous 3D coordinate. The surface location is then determined by marching along each ray with a differentiable rendering function consisting of a recurrent neural network. However, these methods are often limited to low geometric details, leading to over-smoothed results with less high-frequency content. Some works [10], [21] represent the scene as a collection of parametric surface elements and deform the surface via the neural network with a shape prior. Lin *et al.* [21] further introduce multi-view photometric consistency with a piecewise image alignment strategy to reconstruct shape parameters from videos without any depth or mask information. However, the above surface representation provides geometry information without texture and illumination, which is

hard to generalize in outdoor scenes with complex illuminations. To relax the requirement of 3D supervision, several works propose to formulate differentiable rendering functions using only 2D images as supervision. Some of them represent surfaces as an implicit field [11], [12], [13] and find the surface intersection for each ray using a numerical method, then calculate an analytic derivative from the predicted colors by differentiable rendering. Others like Nerf [14], [20], [67] represent the scene as transmittance fields, so that the opacity and radiance can be learned from rendered images by the accumulation of multiple points on each ray. However, these methods are either limited to indoor scenes with over-smoothed results estimated by the implicit surface or generate rugged and mistaken surfaces from the transmittance fields.

3 SURFACE RADIANCE FIELD REPRESENTATION

Given a set of images with calibrated cameras, unsupervised MVS reconstructs high-fidelity surfaces with both color and geometry without any 3D supervision. Different from other methods that treat the MVS task as a matching problem, our learning objective is reconstructive: namely, the scene is a combination of 3D shapes, texture, material, illumination, and view directions.

As shown in Fig. 2, all the information is represented locally on a set of surfaces. Unlike the height field that only encodes the relative displacement in one direction, we use a neural network to implicitly represent the geometry as a per-view depth image for each triangle from every viewing direction. Specifically, we assign each of the local surfaces with a geometry and texture embedding, and the geometry MLP encodes all the view-dependent distance information. The distance to the intersection through the target surface of each ray acts as the surface deformation along view-dependent rays. The geometry embedding is first passed through a multilayer perceptron network (MLP) to predict that distance. Then, we predict the color by another MLP that takes the texture embedding, the predicted intersection coordinate, and the view direction as input. This texture MLP can be seen as an enhancement of the surface light field[68], [69], which generates the local radiance field of the texture on the manifold instead of the tangent plane of the potential surface. A detailed demonstration of this improvement can be found in Section 5.3.

To learn such a decomposition without supervision, we build an auto-decoder pipeline that the embedding is optimized through a differentiable rendering framework. By comparing the rendered images with a set of target images, the accurate surface location and color are learned gradually through back-propagation since the points on the true underlying scene surface can generate more accurate colors and lower color differences across multiple views. As shown in Fig. 6, starting from the very sparse points prediction via the Structure-from-Motion (SfM), we progressively refine the underlying surface structures by coarse-to-fine points sampling.

The following sections describe how this is done. We introduce the surface radiance field in Section 3.1, followed by two sampling strategies (Section 3.2) for point cloud generation and view synthesis.

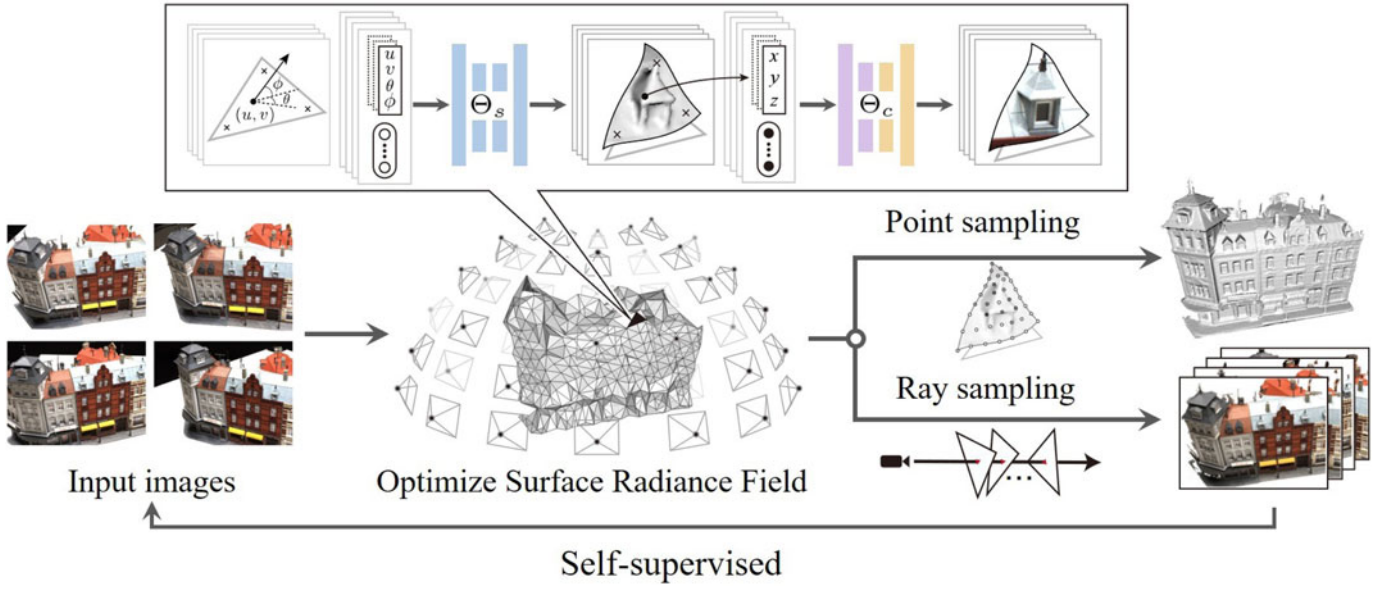


Fig. 2. An overview of our Surface Radiance Field (SurRF) and unsupervised optimization procedure. We represent the scene on a set of surfaces where each of them is a combination of 3D shapes, textures, and view directions. We optimize it through a differentiable rendering framework by comparing the rendered images with a set of target images.

3.1 Definition

The key idea is to represent the scene from a set of surfaces $\mathbf{S}^*$ with continuous geometry and texture information. Each surface $S_i^* \in \mathbf{S}^*$ is a triangle facet containing two learnable embeddings γ_i^s and γ_i^c with code length l_s and l_c to describe the surface geometry and texture respectively. We use a multilayer perceptron network (MLP) to encode the surface indexed by the texture coordinate $\psi^{(x)} = (u, v)$ and the ray direction $\psi^{(d)} = (\theta, \phi)$. So given a ray whose normalized direction is d_i^* hitting at the 3D point x_i^* of the triangle facet, the surface deformation is given by a residual prediction:

$$s = x_i^* + r_{\Theta_s}(\psi^{(x)}, \psi^{(d)}; \gamma_i^s) d_i^*, \quad (1)$$

where

$$r_{\Theta_s} : R^2 \times R^2 \times R^{l_s} \rightarrow R, \quad (2)$$

encodes the residual length at x_i^* in view direction d_i^* from the beginning point on the triangle facet to the target surface.

We further predict the color as a function of both the estimated 3D location of the target surface and the view direction:

$$C_{\Theta_c} : R^3 \times R^2 \times R^{l_c} \rightarrow R^3 \quad (3)$$

$$(s - \bar{x}_i^*, \psi^{(d)}; \gamma_i^c) \rightarrow (c)$$

where $\bar{x}_i^*$ is the centroid of the triangle. It is worth noting that unlike the geometry MLP that maps from the point (2D) and the direction (2D) on the primary triangle facet, the texture MLP maps from the 3D point, followed by the texture fields [60]. The different choices of input coordinate of the geometry and texture MLP enable the view-consistent 3D geometry. If the geometry MLP produces inconsistent 3D geometry, i.e., different 3D points prediction on different view directions, the texture MLP will generate inconsistent color values based on the inconsistent 3D points predictions. It is exactly that the inconsistency constrains the differentiable

rendering process to produce photometric consistent output. A more detailed discussion can be found in Section 5.3.

In general, our full representation can be written as:

$$L_{\Theta} : R^4 \times R^l \rightarrow R^3 \times R^3 \quad (4)$$

$$(\psi_i, \gamma_i) \rightarrow (s, c)$$

where $\psi_i = (\psi_i^{(x)}, \psi_i^{(d)})$, $\gamma_i = (\gamma_i^s, \gamma_i^c)$ and $l = l_s + l_c$.

3.2 Sampling

The goal during sampling is to compose global scene geometry and color from a collection of surface radiance fields. As illustrated in Fig. 4, the sampled objects can be either surface points, used for point cloud generation; or pixel rays from different camera poses, used for view synthesis and unsupervised learning.

Point Cloud Generation. As shown in Fig. 4a, the key design in point cloud generation is to sample a height field

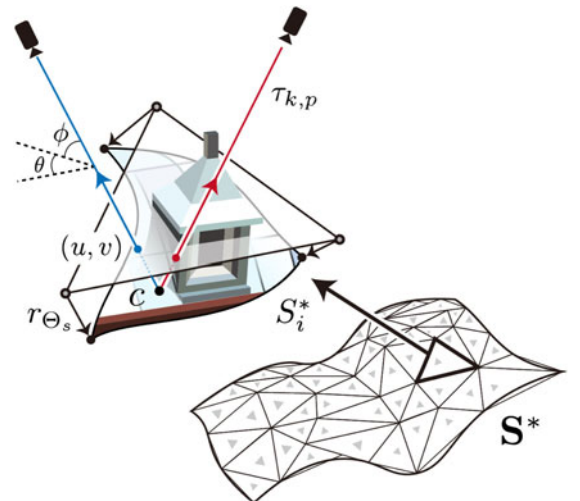


Fig. 3. Illustration of our Surface Radiance Field (SurRF) representation.

TABLE 1
Mathematical Notation Guide

Symbol	Meaning
$\gamma_i^s, \gamma_i^c, \gamma_i$	Geometry embedding, texture embedding, and their concatenation.
$r_{\Theta_s}, C_{\Theta_c}, B_{\Theta_b}, L_{\Theta}$	Networks of deformation, color, lighting and their combination.
l_s, l_c, l	Length of geometry, texture and their concatenation embedding.
$\psi^{(x)}, \psi^{(d)}, \psi_i$	Texture coordinate, ray direction and their combination.
x_i^*, d_i^*	3D point and the normalized direction at the triangle facet.
$\bar{x}_i^*, n$	Centroid and normal of the triangle.
$s, c^{(a)}, c^{(b)}, c^{(m)}, c$	Predicted surface location, view-independent color, view-dependent color, specular material and the total color.
γ^b, l_b	Lighting embedding and its length.
$T(a), P_i(a), P(a)$	Sampling coordinates, the generated points set and the final point cloud.
$t_k, \tau_{k,p}, \psi_{i,k,p}$	Camera center, the ray through pixel p , and the intersection at the triangle facet.
$c_{i,k,p}^{(a)}, c_{i,k,p}^{(b)}, c_{i,k,p}$	Predicted view-independent, view-dependent and the total color at pixel p .
$\bar{s}_{k,p}, s_{k,p}^{(a)}, \bar{c}_{k,p}, c_{k,p}^{(a)}$	Mean and variance of the predicted location and color at each ray.

on each local triangle facet. Since our geometry is a per-view depth image from every view direction, theoretically, we can select any direction to generate the corresponding distance to the intersection point. In practice, we use the surface normal as the sampling view direction for convenience. Besides, it prevents the network from singular or over-extrapolated coordinate inputs. For each surface, we first sample a set of texture coordinates $T(a)$ as:

$$T(a) = \left\{ \left(u, v, 0, \frac{\pi}{2} \right) \mid u, v \in \left\{ 0, \frac{1}{a}, \frac{2}{a}, \dots, 1 \right\}, u + v \leq 1 \right\}, \quad (5)$$

where $a \in N^*$ is the number of samples on one side. For each surface, the generated points set P_i is:

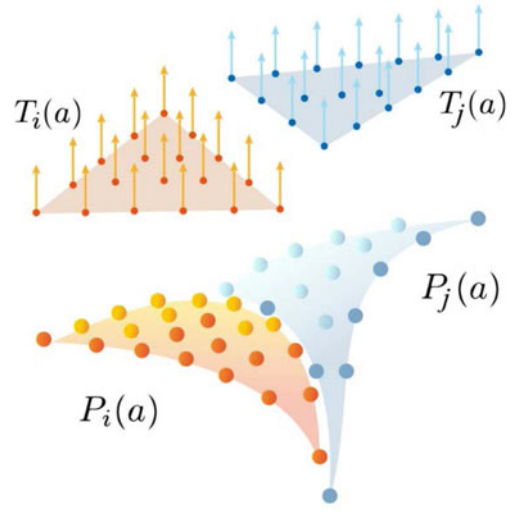
$$P_i(a) = \{(s, c) \mid (s, c) = L_{\Theta}(\psi_i, \gamma_i), \psi_i \in T(a)\}, \quad (6)$$

and the final point cloud is the union: $P(a) = \bigcup_i P_i(a)$.

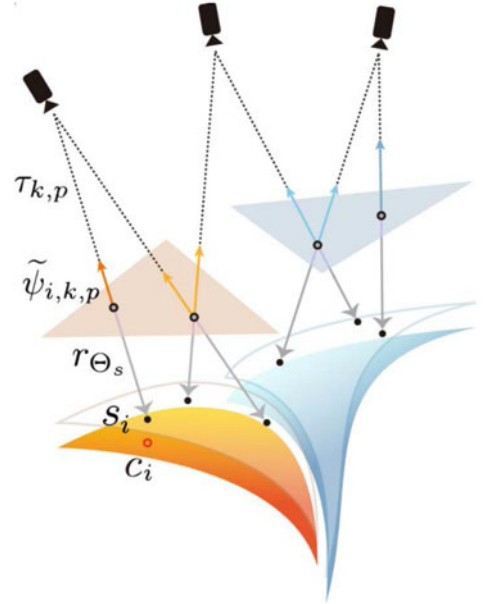
View Synthesis. The main part for view synthesis is a differentiable renderer like the soft rasterizer [70], where we predict the deformation and color, as shown in Fig. 4b, and aggregate these predictions globally utilizing z-buffer in the traditional point-based graphics, as shown in Fig. 2. Specifically, given images from K views with camera centers t_k , our goal is to generate the color per pixel.

For each pixel p , the ray $\tau_{k,p}$ through it intersects the triangle facet at $\psi_{i,k,p}$. We only select the triangle facet whose intersection is inside itself. Since a ray may pass through a complex surface several times, we only collect N_f closest to prevent unnecessary sampling during optimization. These selected facets are marked as $D_{k,p}$. The sampling number N_f is only related to the training efficiency and has nothing to do with the performance. Then the rendered pixel color is the average value of the selected surfaces:

$$(\bar{s}_{k,p}, \bar{c}_{k,p}) = \frac{\sum_i \mathbf{1}_{i \in D_{k,p}} L_{\Theta}(\psi_{i,k,p}, \gamma_i)}{\sum_i \mathbf{1}_{i \in D_{k,p}}}. \quad (7)$$



(a) Point sampling for point cloud generation.



(b) Ray sampling for view synthesis.

Fig. 4. Illustration of different sampling strategies to compose global scene geometry and images.

4 DETAILED SURFACE RECONSTRUCTION

In the previous section, we have described the essence of our scene representation. However, the direct way for unsupervised training that compares the colors sampled from the surface radiance field with the target image fails to achieve state-of-the-art quality. We introduce the other two components to enable learning high-accuracy and continuous surfaces in complex scenes. First, we additionally model different appearances such as surface materials and lighting (Section 4.1), which enforces detailed geometry learned from view-dependent colored rays. The second is a shape continuity loss (in Section 4.2) to guarantee a smooth and continuous surface prediction globally.

4.1 Rendering With Lighting Approximation

Although the new representation enables to encode various colors in different view directions, the direct optimization

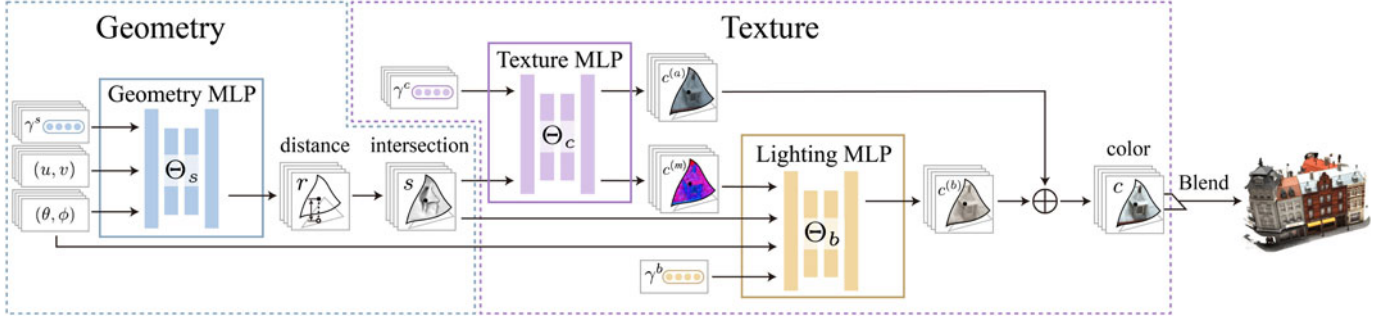


Fig. 5. The network architecture of our proposed SurRF. The surface geometry is first predicted using the 2D texture coordinate, ray direction and an embedding through a geometry multilayer perceptron network (MLP). Then, we model the illumination and estimate the surface color through another two networks modeling texture and lighting.

on it failed to find either the geometry prior like surface normal and multi-view consistency, or the view-independent materials such as ambient and diffuse lighting.

The detailed network architecture of our proposed SurRF is shown in Fig. 5, where we locally represent the geometry and texture of the scene, and globally encode the illumination. To facilitate the modeling of illumination, we propose to improve the color prediction C_{Θ_c} in Eq. (3) with an enhanced function, together with a render equation B_{Θ_b} :

$$C_{\Theta_c} : R^3 \times R^{l_c} \rightarrow R^3 \times R^6 \quad (8)$$

$$(s - \bar{x}_i^*; \gamma_i^c) \rightarrow (c^{(a)}; c^{(m)}),$$

$$B_{\Theta_b} : R^8 \times R^6 \times R^{l_b} \rightarrow R^3 \quad (9)$$

$$(s, \psi^{(d)}, n; c^{(m)}; \gamma^b) \rightarrow (c^{(b)})$$

where $c^{(a)}$ is the view-independent color, and the view-dependent color $c^{(b)}$ is generated by the diffuse and specular material $c^{(m)}$, together with the surface normal n computed from the initial triangle facet, predicted surface location s , view direction $\psi^{(d)}$ and a latent lighting embedding γ^b with length l_b . The reason why we use s instead of $s - \bar{x}_i^*$ in Eq. (9) is that the learning targets of the texture embedding and the lighting embedding are different. While the texture embedding tries to recover local color and materials, the lighting embedding aims to model the rendering equations like BRDF [71], [72], which is a combination of the surface geometry, material and illumination parameters. As shown in Fig. 5 and Eq. (9), unlike the texture representation that assigns each facet with a unique embedding γ_i^c , all the facets share with the same lighting embedding γ^b .

The final color from camera k at pixel p is given by:

$$c_{i,k,p} = c_{i,k,p}^{(a)} + c_{i,k,p}^{(b)}. \quad (10)$$

4.2 Optimization

To enable learning high-resolution continuous surfaces in complex scenes, we adopt two additional strategies during ray sampling.

Shape Continuity. The deformed surface along each ray encoded by a set of separate surface radiance fields may lead to inconsistent estimation at the edge of each surface. The direct way to reduce discontinuity by decreasing the distance with adjacent surfaces is onerous because the latent embedding of each surface can only be optimized by sampled rays,

which requires heavy computation to judge connectivity relationships. Unlike the previous fusion methods [70], [73] that ignores the differences along each ray, our solution further computes and reduces the variance of the surface location and color $(s_{k,p}^{(\sigma)}, c_{k,p}^{(\sigma)})$. Followed by Eq. (7), we have:

$$(s_{k,p}^{(\sigma)}, c_{k,p}^{(\sigma)}) = \frac{\sum_i \mathbf{1}_{i \in D_{k,p}} [L_{\Theta}(\psi_{i,k,p}, \gamma_i) - (\bar{s}_{k,p}, \bar{c}_{k,p})]^2}{\sum_i \mathbf{1}_{i \in D_{k,p}}}. \quad (11)$$

Note that this shape consistency optimization follows the same computation pipeline with view synthesis, which requires no additional sampling. Since different camera positions and picture sizes will change the local coordinates on each surface, the continuity is guaranteed by sufficient training samples.

Progressive Training. The points sampling strategy mentioned in Section 3.2 enables us to progressively refine triangle facets of each surface to obtain more complex topology and detailed self-occlusion information. After generating the previous stage prediction S^* , we sample a set of detailed points $P(a)$. At the following stage, detailed triangle facets are obtained from either ball pivoting algorithm[74], or use an equilateral triangle with an appropriate side length whose centroid coincides with the point and three sides perpendicular to the normal of that point. Before the triangle mesh generation procedure, we resample the point cloud to attain uniformly sampled points with resolution v . The resampling rule is as follows: The scene space is divided by voxels with size v . If multiple points fall into the same voxel, we only preserve the mean of these points. At the following stage, the point cloud resolution is two times finer than the previous stage. The triangle facets at the first five stages are obtained from the ball pivoting algorithm, with a radius of $2v$. At the final stage, we use an equilateral triangle with a side length of $1.5v$. The reason to use the equilateral triangle is for convenience because when the number of points becomes large, it is difficult and time-consuming for the ball pivoting algorithm to generate a set of triangle facets without holes among them.

The network structure is consistent with the previous stage and inherits the network parameters Θ to accelerate training. In practice, we use a 6 stage prediction with the number of samples $a = 4$. The number of samples a at the finest stage is set to 10 for the final point cloud generation. An illustration of the progressive training is shown in Fig. 6.

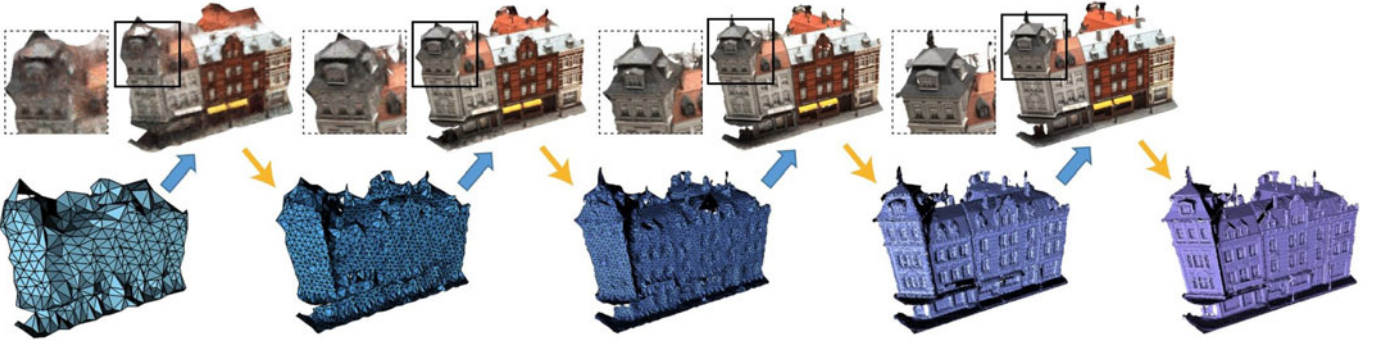


Fig. 6. Illustration of progressive training. Our method can progressively refine underlying surface structures via coarse-to-fine points sampling from only a set of posed RGB images.

4.3 Implementation Details

We trained for each scene separately by sampling pixel rays from a batch of randomly selected camera views. Our loss function is composed with the rendering differences and shape continuity, defined as:

$$Loss_c = \sum_{k \in V_k} \sum_p \|\tilde{c}_{k,p} - I_{k,p}\|_2^2, \quad (12)$$

$$Loss_s = \sum_{k \in V_k} \sum_p \left(\|c_{k,p}^{(\sigma)}\|_2^2 + \beta \cdot \|s_{k,p}^{(\sigma)}\|_2^2 \right) \quad (13)$$

where V_k is the randomly selected views, and the total loss is given by:

$$Loss = Loss_c + \alpha \cdot Loss_s. \quad (14)$$

In our experiments, $\alpha = 10.0$ and $\beta = 0.01$. During training, the number of views rendered at one time is $\|V_k\| = 4$, with the random image scale from 500×500 to 600×600 . Each of the prediction function in $(r_{\theta_s}, C_{\theta_c}, B_{\theta_b})$ is an MLP with 4 fully-connected layers. Followed by SIREN[75], all activation functions are set to sine. The channels per layer keeps the same with the input code length, where $l_s = l_c = 32, l_b = 16$. We use the Adam optimizer[76] with an initial learning rate of 1.0×10^{-4} which is decreased by 0.9 for every 100 epochs. The predictions at the first five stages in progressive training are trained alone for 5000 epochs, with $N_f = 5$. At the final stage, four views are selected from the input images as the validation set. These views are evenly distributed in the range of all view directions in the dataset. Besides, due to the memory limitation, each picture is divided into several small blocks. Each block has an image resolution close to 500×500 . Then, the learning rate of the network is set to 1.0×10^{-5} and train until both the rendering loss $Loss_c$ and the shape continuity loss $Loss_s$ no longer drops. At the final stage, $N_f = 3$. All the training is conducted on a single GTX 1080Ti graphics card.

5 EXPERIMENT

We first demonstrate the state-of-the-art unsupervised-MVS performance of our proposed SurRF on the DTU[77] and Tanks and Temples[18] datasets. Our method ranks the 1st in DTU and Tanks and Temples benchmark among all learning-based unsupervised methods.

5.1 DTU Dataset

We evaluate our method on the DTU evaluation dataset [77], which is a large-scale MVS dataset consisting of 124 different scenes scanned at 49 or 64 camera positions under 7 different lighting conditions. The dataset is split into training, validation and evaluation sets [5].

Quantitative results are shown in Table 2, where the accuracy and completeness of the final reconstructions are calculated using the distance metric [77]. The overall distance metric is measured as the average of accuracy and completeness. While Gipuma[1] performs the best accuracy, our method outperforms all unsupervised methods in completeness and overall score, and performs competitive results over state-of-the-art supervised methods in all distance metrics. The qualitative comparison including the depth visualization of our method and other methods [7], [8], [17] on five scans is shown in Fig. 7. Our method generates smoother point clouds and more continuous surfaces with finer details. In particular, we demonstrate superior fidelity in terms of shape and color.

5.2 Tanks and Temples Dataset

The Tanks and Temples dataset [18] contains real-world large scenes under complex lighting conditions. The corresponding quantitative results on the intermediate dataset

TABLE 2
Quantitative Results of Reconstruction Quality in Terms of the Distance Metric (Lower is Better) on the DTU Evaluation Dataset[77]

Method		Acc.(mm)	Comp.(mm)	Overall(mm)
Supervised	MVSNet[3]	0.396	0.527	0.462
	SurfaceNet+[5]	0.385	0.448	0.416
	Point-MVSNet[7]	0.342	0.441	0.376
	CasMVSNet[16]	0.325	0.385	0.355
	UCS-Net[17]	0.338	0.349	0.344
	PVA-MVSNet[51]	0.379	0.336	0.357
	D ² HC-RMVSNet[78]	0.395	0.378	0.386
Unsupervised	Furu[23]	0.613	0.941	0.777
	Tola[24]	0.342	1.190	0.766
	Gipuma[1]	0.283	0.873	0.578
	COLMAP[15]	0.400	0.664	0.532
	Robust Consistency [8]	0.881	1.073	0.977
	SurRF(Ours)	0.388	0.390	0.389

Our method, SurRF, outperforms state-of-the-art unsupervised methods in completeness and overall score, and performs competitive results over state-of-the-art supervised methods in all evaluation metrics.

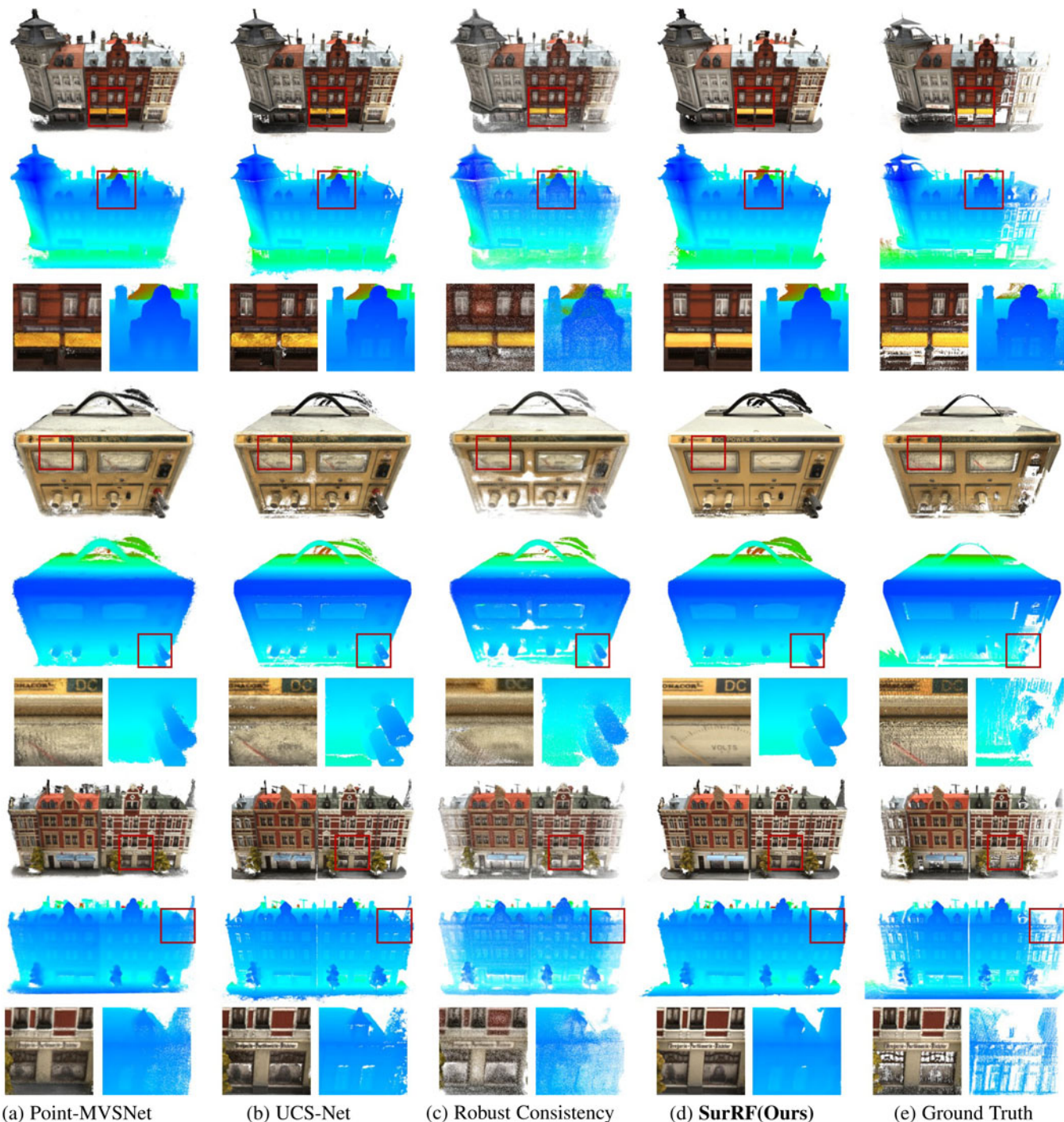


Fig. 7. Qualitative results of scan 9, 11, and 15 in the DTU dataset[77] compared with Point-MVSNet[7], UCS-Net[17], Robust Consistency[8] and ground truth point cloud. Note that our proposed SurRF generates denser point clouds with better continuity and colors. For fair comparison, we generate each result in higher completeness.

are denoted in Table 3. Our proposed SurRF achieves state-of-the-art performance among the unsupervised multi-view stereo methods and competitive results over state-of-the-art supervised methods. Specifically, the performance of our method is significantly higher than all the unsupervised learning-based MVS methods with a big margin. The reconstructed point clouds are shown in Fig. 8, compared with COLMAP[15], R-MVSNet[4], CasMVSNet[16] and UCS-Net [17]. With a competitive F-score value over state-of-the-art supervised methods, our SurRF can produce much more complete, delicate, and realistic surface predictions.

5.3 Ablation Studies

In this section, we provide ablation experiments to analyze the strengths of the key components of SurRF. Experiments are performed and evaluated on a specific model (scan 9) in the DTU dataset. It contains multiple typical challenging cases such as complex surface topology, repeating patterns, and various illuminations. For a fair comparison, all the predictions were started from the same initial surfaces with an average resolution of 4.0 mm, which is ten times coarser than the required resolution provided by the ground-truth. This initialization was generated by the first four-stage prediction

TABLE 3
Quantitative Results of State-of-the-Art Learning-Based Methods on the Tanks and Temples (T&T) Dataset [79] Leaderboard

Method		Rank	Mean	Family	Francis	Horse	Lighthouse	M60	Panther	Playground	Train
Learning based Supervised	VisMVSNet [80]	8.12	60.03	77.40	60.23	47.07	63.44	62.21	57.28	60.54	52.07
	D ² HC-RMVSNet [78]	9.50	59.20	74.69	56.04	49.42	60.08	59.81	59.61	60.04	53.92
	CasMVSNet [16]	23.38	56.84	76.37	58.45	46.26	55.81	56.11	54.06	58.18	49.51
	UCSNet [17]	34.50	54.83	76.09	53.16	43.03	54.00	55.60	51.49	57.38	47.89
	PVAMVSNet [51]	34.88	54.46	69.36	46.80	46.01	55.74	57.23	54.75	56.70	49.06
	SurfaceNet+[6]	51.38	49.38	62.38	32.35	29.35	62.86	54.77	54.14	56.13	43.10
	Dense R-MVSNet [4]	57.75	50.55	73.01	54.46	43.42	43.88	46.80	46.69	50.87	45.25
	VisibilityAwarePointMVSNet [81]	62.62	48.70	61.95	43.73	34.45	50.01	52.67	49.71	52.29	44.75
	Point-MVSNet [7]	62.75	48.27	61.79	41.15	34.20	50.79	51.97	50.85	52.38	43.06
Learning based Unsupervised	MVSNet [3]	78.50	43.48	55.99	28.55	25.07	50.79	53.96	50.86	47.90	34.69
	MVS ² [9]	100.38	37.21	47.74	21.55	19.50	44.54	44.86	46.32	43.48	29.72
	M ³ VSNNet [53]	100.38	37.67	47.74	24.38	18.74	44.42	43.45	44.95	47.39	30.31
	SurRF(Ours)	31.88	54.36	69.69	52.45	27.83	63.08	59.41	52.33	59.34	50.74

Our proposed SurRF achieves state-of-the-art performance among the unsupervised MVS methods and competitive results over state-of-the-art supervised methods.

of the progressive training as described in Section 4.2. At each stage we train our model with 10k iterations with a fixed learning rate of 1.0×10^{-5} . In these experiments, the rendered images for training and evaluation were all down-scaled to 600×800 from the original size of 1200×1600 . To measure the visual effect, we compare the photos with the rendered results under specific views using PSNR. Besides, a completeness mask is added that marks pixels excluded from the boundary of the reconstruction. We qualitatively and quantitatively illustrate the performance gain of each component in Fig. 9 and Table 4.

Surface Residual Prediction. By disabling the surface residual prediction r_{Θ_s} , the surface coordinate in Eq. (1) changed to: $s = x_i^*$. This degradation in geometry can be seen as a special form of surface light field[68], [69], which only generates the local radiance field of the texture on the tangent plane of the potential surface. As shown in Table 4, both the performance on geometry (F-score under 1mm) and masked rendered images (PSNR) drops significantly since the points away from the underlying scene surface will generate a mean value of colors from nearby surfaces through different view directions. As shown in Fig. 9a, compared with the full model in Fig. 9d, this setting will generate blurred rendered textures, which illustrates the importance of residual prediction for detailed multi-view synthesis.

Lighting Approximation. In this part, we disabled lighting approximation (Section 4.1) in the color prediction on each ray. In other words, the color prediction C_{Θ_c} in Eq. (3) was represented by only one MLP. For a fair comparison, the structure of the network was consistent with the original one, which contains four layers of 32-dimensional and four layers of 16-dimensional feature vectors. As shown in Table 4, both the PSNR and F-score drop dramatically. Qualitative results are shown in Fig. 9b, where rough geometry estimation occurred with blurred texture predictions. The lack of geometric modeling for color prediction blocks the backward loss gradient from the rendering errors to the geometry refinement, which further exacerbates the rendered images. This demonstrates the effectiveness and robustness of lighting approximation for accurate surface prediction under shading and reflection regions.

Shape Continuity. In this part, we trained our network without shape continuity loss $Loss_s$, i.e., $\alpha = 0$ in Eq. (14). Table 4 shows that only the F-score drops dramatically with a close PSNR value. Surprisingly, training without shape prior losses leads to high-frequency texture output with over deformed and discontinuous shapes, which is further illustrated in Fig. 9c. Since our continuous surface formulation is only valid in local regions, unsupervised geometry refinement with texture guidance will inevitably lead to globally distorted geometric predictions. Therefore, using the shape continuity loss in the rendering pipeline not only enforces color and shape consistency between the border of nearby surfaces but also provides a geometric prior for surface consistency.

Progressive Training Stage. The quantitative results with different stages in progressive training described in Section 4.2 are summarized in Table 5. After training at each stage, we upsample the point cloud using the strategy described in Section 3.2, enabling the representation to capture more detailed features. The average resolution of the initial mesh at the next stage is two-time finer than the previous stage, resulting in an approximately four-times increase in the number of surfaces. We find that at the initial stages, as the number of surfaces increases, the quality of PSNR and F-score both remarkably increases. At the final stage, while the quality of PSNR stabilizes, the F-score increases dramatically. Qualitative results are shown in Fig. 6.

Choices of Coordinates. In Table 6, we show the ablation experiments to analyze the choices of coordinates of the texture MLP, where 3D global means the color module takes s , 3D local means the color module takes $s - \bar{x}_i^*$ and 4D means the color module takes (u, v, θ, ϕ) . As shown in Table 6, changing the color module (from the 3D local coordinate $s - \bar{x}_i^*$ to the 3D global coordinate s) does not affect the performance. However, when changing to the 4D coordinate (u, v, θ, ϕ) , F-score and PSNR drop largely. As a result, mapping from the 3D point in the texture MLP does help predict accurate geometry, i.e., help produce consistent 3D geometry. We further show the ablation experiments to analyze the choices of coordinates of the lighting MLP. As shown in Table 6, when changing from the 3D global coordinate s to the 3D local coordinate $s - \bar{x}_i^*$ in the texture MLP, F-score

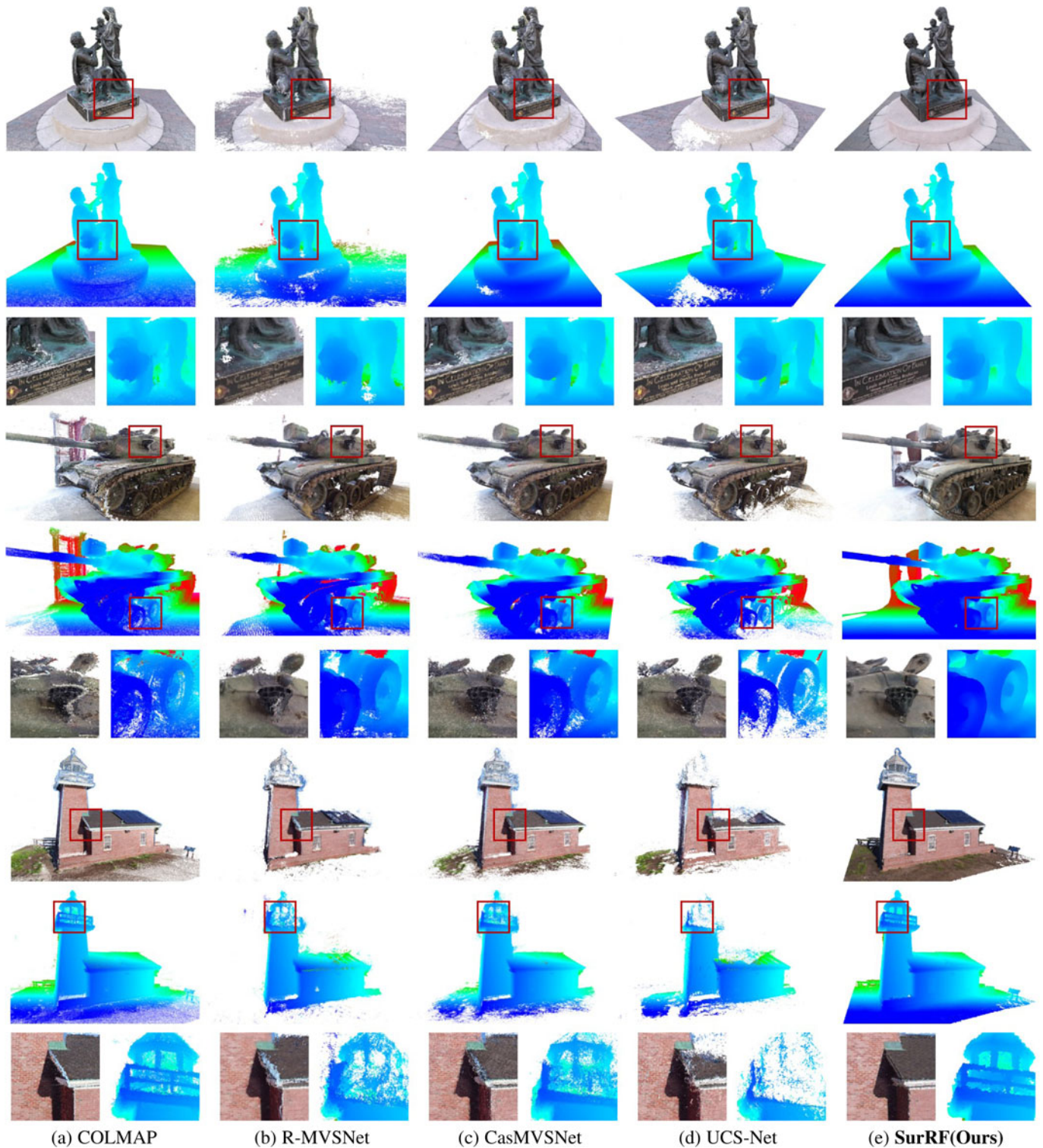


Fig. 8. Point cloud and depth visualization of Family, M60, and Lighthouse in the intermediate set of Tanks and Temples dataset[18] compared with COLMAP[15], R-MVSNet[4], CasMVSNet[16] and UCS-Net[17]. Our proposed SurRF produces much more complete, delicate, continuous, and realistic surface predictions.

and PSNR drop largely. As a result, the illumination should be encoded globally.

5.4 Editing

Since the representation of SurRF both has merits in the unstructured point cloud globally and radiance field locally, it is convenient to edit the reconstruction like a mesh such as

scene manipulation and relighting. We used a five-stage progressive training on scan 9 of the DTU dataset and manipulate the mesh location of the final stage, as shown in Fig. 11. Pay attention that the image c described in Section 4.1 is defined as the summary of the view-independent reflectance $c(a)$ and the view-dependent shading $c(b)$, i.e., $\text{image} = \text{reflectance} + \text{shading}$ [82], which is different from

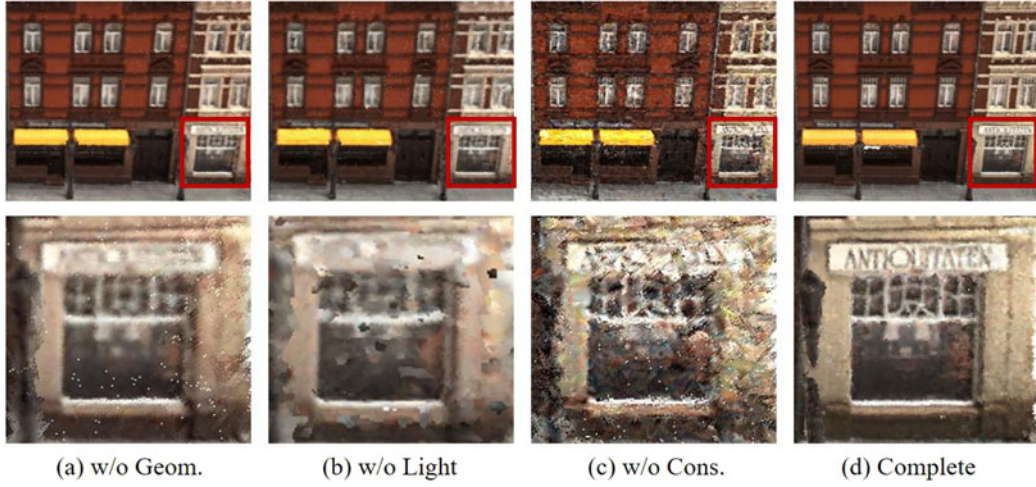


Fig. 9. Qualitative results of ablation study on different components. Geom. denotes Surface residual prediction, Light denotes Lighting approximation and Cons. denotes Shape continuity loss.

the intrinsic decomposition model, i.e., image = reflectance $\odot$ shading. Therefore, in our case, the shading component in the relighting experiment does not look like the traditional model. Nonetheless, the learned surface shapes, albedo, and materials can still be readily used to synthesize images from unseen lighting conditions. In Fig. 10, the first line of shading was learned from our representation, and we simply generate the new relighting results with the classic Phong shading.

5.5 Compactness

Since SurRF composes each local surface as a 2D manifold with latent codes, this representation naturally enjoys advantages in storage and compression.

Network Compactness. In Table 7, we report the quantitative comparison among methods (Nerf [14], IDR [13], and the proposed SurRF) from three aspects: 1) the time efficiency

for training and rendering, 2) the memory efficiency in terms of the GPU memory occupancy and the network capacity, and 3) the geometric reconstruction performance. Both the time efficiency for rendering and the memory efficiency are evaluated by sampling 10,000 rays. The total ray number during the optimization is around 20 million rays per scene from 49 views. Since the selected test models of the DTU dataset in the IDR evaluation is not consistent with the state-of-the-art methods [7], [8], [9], and their released code requires careful parameter tuning and foreground mask as extra inputs, we only report the results on their selected test models (24, 37, 40, 55, 63, 65, 69, 83, 97, 105, 106, 110, 114, 118 and 122) for a fair comparison.

TABLE 6
Ablation Study on Different Choices of Coordinates

Texture	Lighting	F-score(<1mm)	PSNR
3D global	3D global	89.79	23.67
4D		63.21	21.34
3D local	3D local	61.47	19.45
3D local	3D global	89.82	23.76

TABLE 4
Ablation Study on Different Components, Which Demonstrates the Effectiveness of Surface Residual Prediction (Geom.), Lighting Approximation (Light) and Shape Continuity Loss (Cons.)

Geom.	Light	Cons.	F-score(<1mm)	PSNR
✓	✓	✓	89.82	23.76
×	✓	✓	66.73	21.10
✓	×	✓	58.01	18.81
✓	✓	×	77.19	17.73

TABLE 5
Ablation Study on Progressive Refinement on the DTU Evaluation Dataset, Which Demonstrates the Necessity of Progressive Training

Stage	No. of surfaces	F-score(<1mm)	PSNR
1	410	2.56	13.2082
2	1621	23.61	17.1311
3	5419	31.55	18.3671
4	20587	56.62	22.1507
5	77777	75.45	23.6537
final	203715	91.63	23.97

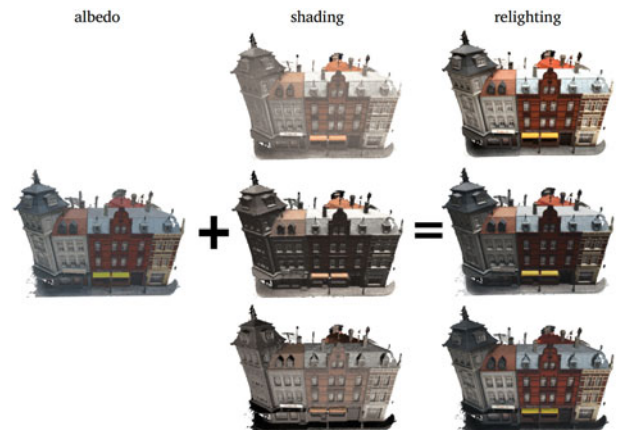


Fig. 10. Relighting on scan 9 in DTU dataset. We generate the albedo, materials, and shadings from our method and only relight the shading with unseen illumination.

TABLE 7
Comparison With Nerf [20] and IDR [13]

Method		Time efficiency		Memory efficiency (10,000 rays)		Geometric measurement
		Optimization ($\sim 2 \times 10^7$ rays) (h)	Rendering (10,000 rays) (s)	No. of parameters	GPU Mem. (MB)	
SurRF	First 5 stages	20.41	0.01	369,078	442	0.88
	Final stage	10.05	0.03		503	0.62
Nerf [20]		24.31	0.29		17,165	1.49
IDR [13]		16.75	0.36	2,915,258	4,196	0.9

The table reports the average results on 15 models used in the IDR evaluation set.

In Fig. 12, we show the rendering and depth visualizations of scan 24 in the DTU dataset, compared with Nerf and IDR, regarding both the ground truth image (above) and the point cloud (below). It can be shown that Nerf and IDR tend to generate more blurry rendering results. Besides, Nerf fails to predict an accurate depth map since it represents the geometry as an opacity field in the dense volume. IDR, on the other hand, tends to generate over-smoothed surfaces, especially at the edges of objects with sharp corners. As shown in Table 7, the proposed method, SurRF, outperforms Nerf and IDR under the Chamfer distance. Besides, SurRF shows more than 10x rendering speedup (final stage) and around 10x GPU efficiency. The gain originates from the fact that we learn the radiance field on a piece of 2D deformable surface rather than in a dense volume in 3D space. The learned low-dimensional manifold structure makes the neural function more accurate and effective to represent a complex 3D scene.

Data Compression. We measure different compression rates and corresponding performances on geometry and rendered images. We use scan 4 of the DTU dataset to conduct this experiment which contained $|S^*| = 18855$ surfaces. The generated point cloud $\hat{P}$ has the same resolution for evaluation, with $|\hat{P}| = 201544$. The total compress ratio C_r is the size ratio of $\hat{P}$ and the total embeddings. Since each point in the usual point cloud representation contains 3 doubles for (x, y, z) coordinate, 3 doubles for (n_x, n_y, n_z) normal values and 3 uchars for RGB color, the total compress ratio C_r is:



Fig. 11. Scene editing for view synthesis, which shows great potential in the graphics pipeline.

$$C_r = \frac{N_p \times |\hat{P}|}{(l \times N_e + N_s) \times |S^*|}, \quad (15)$$

where $N_p = 51$, $N_e = 4$ and $N_s = 24$ represents the byte length for each point cloud, surface embedding elements and surface triangle points. $l = l_s + l_c$ is the length for each surface embedding.

As shown in Fig. 14a, we report PSNR and F-score for color and geometry evaluation under different compression ratios. We also estimate the bandwidth of the compression ratio, which gives a 3dB attenuation of PSNR relative to the peak ($C_r = 1$). This bandwidth reaches more than 6 times, showing the great potential in point cloud compression. Reconstruction results under different compression ratios are shown in Fig. 13. To further examine the contribution of different code lengths (l_s, l_c) for geometry and texture, we fixed one of them to the length of 20 and only changed the other, as shown in Figs. 14b and 14c. In Fig. 14b, though the embedding length of texture remains the same, the PSNR value gradually decreases as the compression ratio of geometry increases, meaning that the geometry refinement plays an important role in multi-view rendering. In Fig. 14c, larger texture embedding with fixed geometry embedding provides little gain on both PSNR and F-score, which further shows that larger geometry embedding plays a more important role in both view synthesis and 3D reconstruction.

5.6 Robustness to the Initialization

To evaluate how the noisy initialization affects the performance, we perform two experiments at the previous level prediction (first 5 stages from the progressive training) and the final level prediction (final stage). Experiments are performed

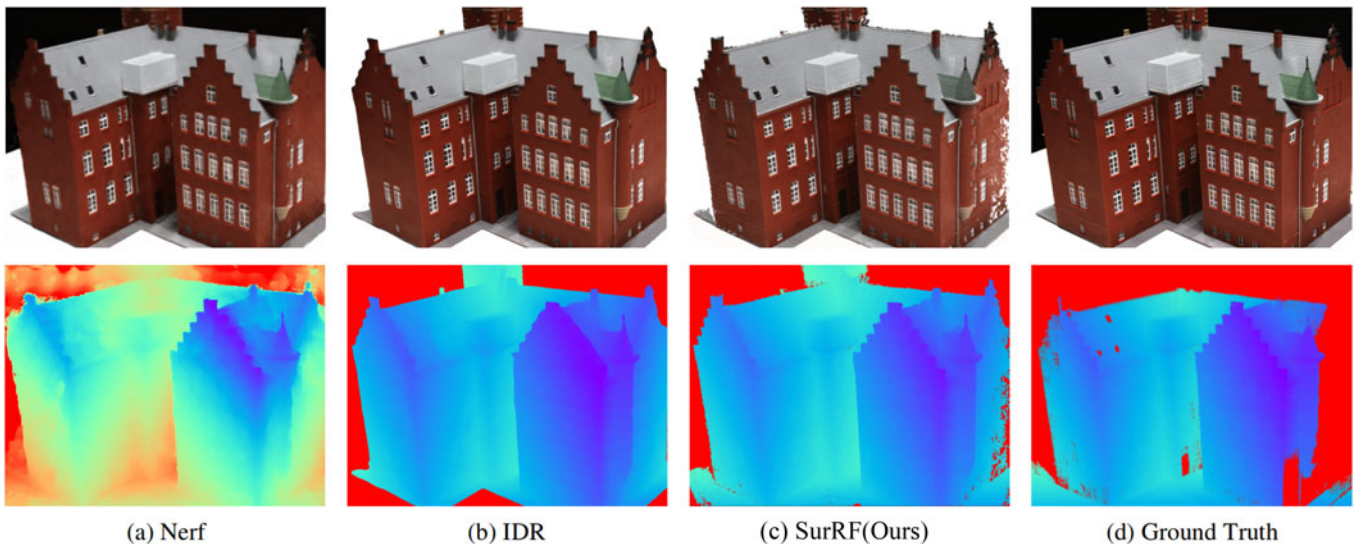


Fig. 12. The rendering and depth visualizations of scan 24 in the DTU dataset, compared with Nerf and IDR, regarding both the ground truth image (above) and the point cloud (below).

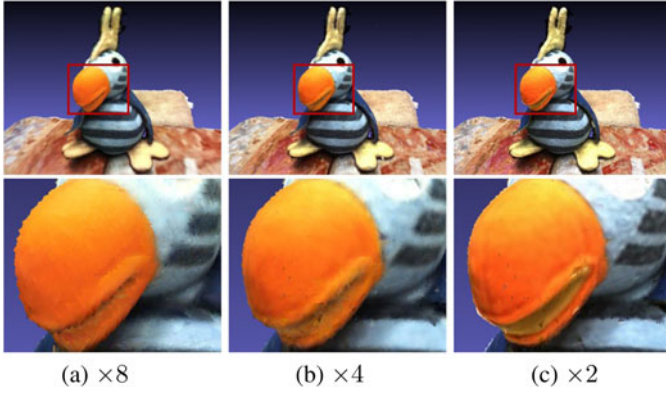
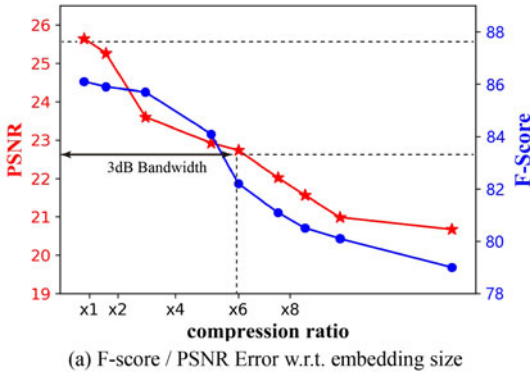
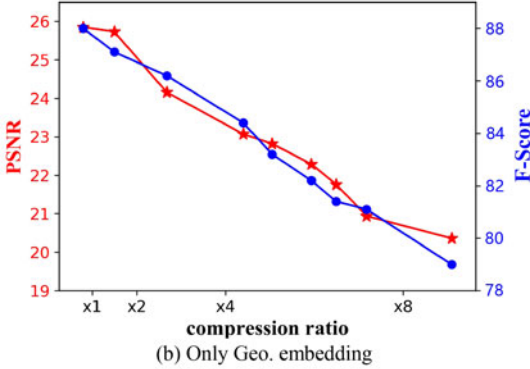


Fig. 13. Qualitative results under different compression ratios.

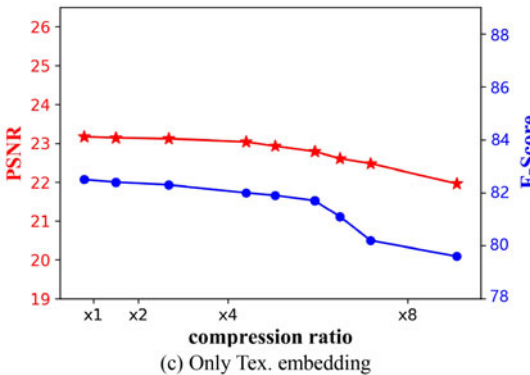
and evaluated on a specific model (scan 9) in the DTU dataset. In Fig. 15, we report the reconstruction performance under different noise levels of triangle facets initialization. Given the point cloud via SfM or the previous reconstruction, we



(a) F-score / PSNR Error w.r.t. embedding size

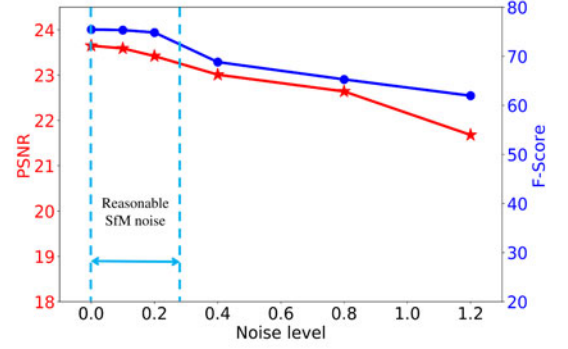


(b) Only Geo. embedding

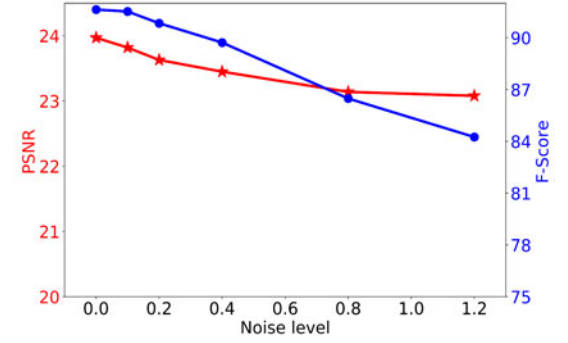


(c) Only Tex. embedding

Fig. 14. Compactness of geometry and texture embedding. We report F-score/PSNR error w.r.t (a) total embedding size, (b) geometry embedding size, and (c) texture embedding size.



(a) First 5 stages of prediction. The triangle facets are initialized from the SfM [83] by COLMAP [15].



(b) Final stage of prediction. The triangle facets are initialized from the previous level point cloud.

Fig. 15. Performance w.r.t. noise level of different triangle facets initialization. The noise level α is defined as a random perturbation $U \sim (-\alpha v, \alpha v)$ adding to each point of the original triangle facets initialization of resolution v .

resample it with resolution v . The noise level is defined as a random perturbation adding to each point. Note that for the SfM [83] from COLMAP [15], when we set the point minimum track length to 4 and the point maximum error (px) up to 20.0 (the default value is 2.0), the noise level is usually smaller than 0.25. However, to explore the performance limit of SurRF, we demonstrate the performance versus the noise level up to 1.2.

As shown in Fig. 15, when the noise level is relatively small ($\alpha \leq 0.4$), the performance of PSNR and F-score remain stable. As the noise level increases ($\alpha > 0.4$), the performance of F-score drops as expected. In particular, the algorithm is more sensitive to the initial triangulation from SfM at the previous

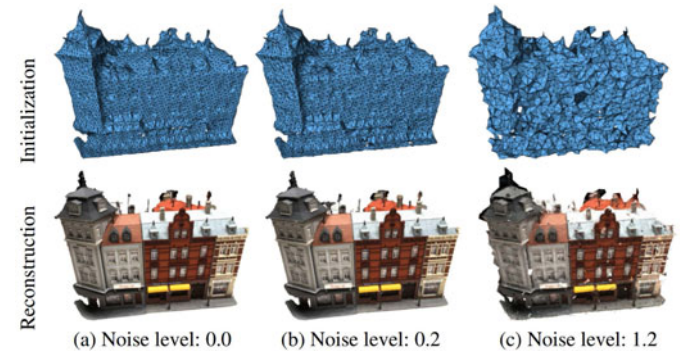


Fig. 16. Visualization of the triangle facets initialization (above) and the reconstruction from the first 5 stages (below) under different noise levels.

TABLE 8
The Comparison of Time Efficiency Using the DTU Dataset

Method		Running time		
		Pre-training	Per-scene optimization	Depth-map extraction(49 views)
Per-scene optimization	SurRF(Ours)	-	29.45 h	73.47 s
	Nerf [20]	-	24.31 h	698.4 s
	IDR [13]	-	16.75 h	855.3 s
Supervised MVS	Cas-MVSNet [16]	-	-	39.984 s
	Point-MVSNet [7]	~1 day	-	249.9 s
	R-MVSNet [4]	-	-	51.45 s

level prediction as shown in Fig. 15a, where the F-score drops from 80 to 60. The reason may lie in the noise amplification effect at multiple stages of predictions. Besides, the noise at the initial stage may lose important structure information, which is difficult to recover from our method. In Fig. 16, we also visualize the initialization of triangle facets (above) and the reconstruction from the first 5 stages (below) under different noise levels. The reconstruction performance remains stable when the noise level is relatively small, while drops when the noise level becomes large, which is consistent with the experiment in Fig. 15. In conclusion, our method is robust in the small noise level of the initial triangulation, while unstable in the large noise level, especially in the previous level of predictions. In the future work, we will additionally consider optimizing the structure of the initial triangle facets (such as the location, the normal and connections between each triangle facet), which would enable SurRF to learn beyond the local feature of the scene.

5.7 Limitation

Supervised MVS methods enjoy fast inference since they take a long time to learn useful structures from a large collection of scenes during training. Per-scene optimization methods, on the other hand, have to infer structure information for each scene. Table 8 shows the running time comparison of the concerned methods (Nerf [20], IDR [13], CasMVSNet [16], Point-MVSNet [7], R-MVSNet [4], and ours). The per-scene optimization time and the depth map extraction time of all methods are compared on scan 9 in the DTU dataset, using 49 views with the size of 600×800 each. There is still much more progress to be made to efficiently optimize the radiance field.

6 CONCLUSION

In this paper, we present a novel unsupervised MVS pipeline by learning Surface Radiance Field (SurRF), a radiance field explicitly defined on the continuous 2D surface. In a local region, SurRF is a continuous representation of both explicit surface deformation and implicit radiance field, encoding geometry, texture, and lighting conditions. A differentiable rendering framework is applied to learn these local SurRFs into a global scene from a particular viewpoint without any 3D supervision. Experiments show outstanding performances in various real-world challenging scenes. Moreover, SurRF shows great potential in owning the joint advantages of mesh (scene manipulation), continuous surface (high geometric resolution), and radiance field (realistic rendering).

ACKNOWLEDGMENTS

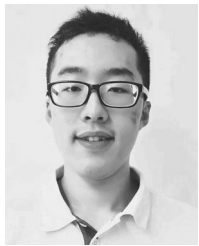
Jinzh Zhang and Mengqi Ji Equal contributed equally to this work.

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